VisitProof

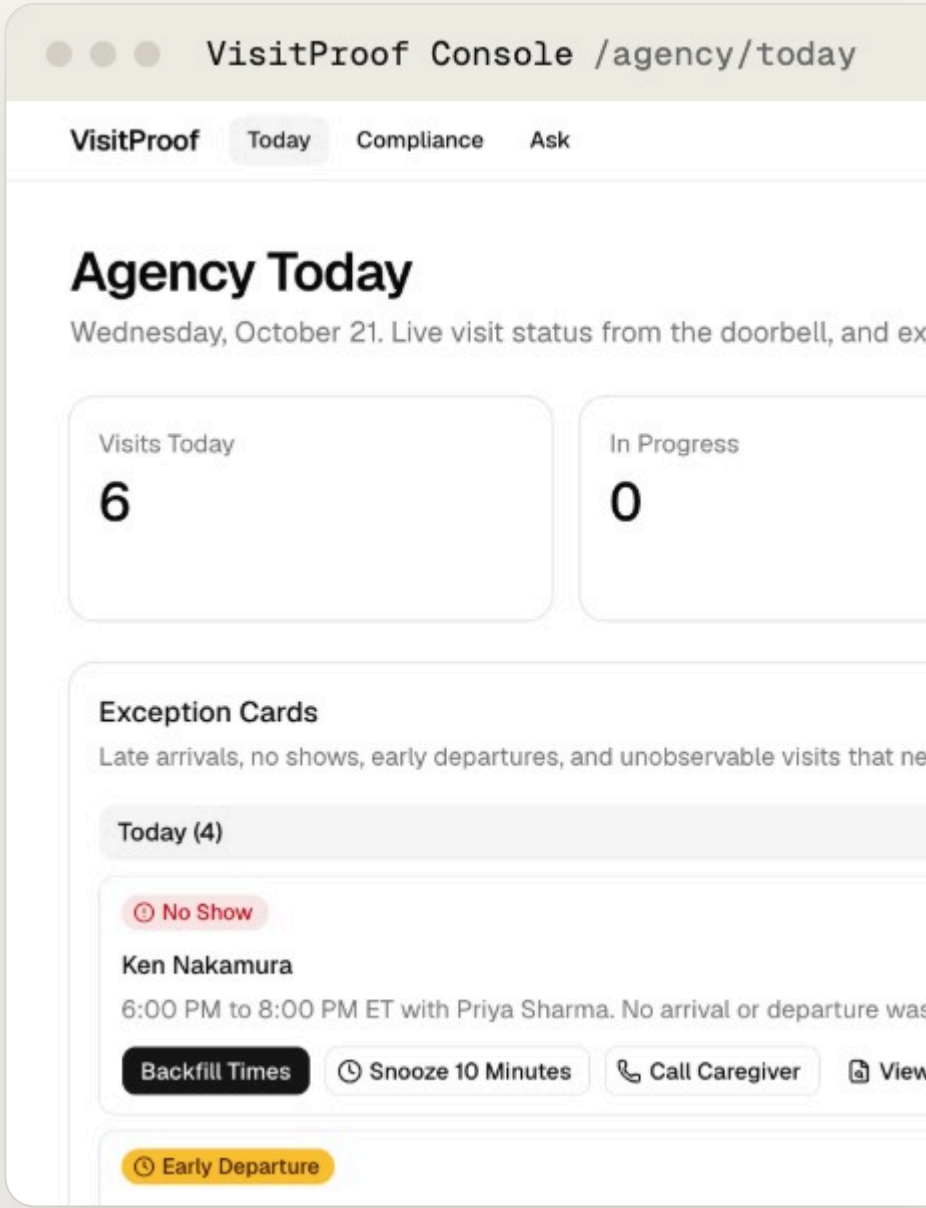
Proof Of Presence From The Doorbell Already At The Door

Real-time no-show alerts, EVV reconciliation, and one calm message a day for families, for Medicaid home care.

JUDGES BRIEF · BUILD, SHIP, SHAPE · RING TRACK

Eugene Mutembei · Technetium Kenya

Live Demo visitproof.codewitheugene.top
Repository github.com/CodeWithEugene/Amazon-Developer-Hackathon
Demo Video Demo video: see the Devpost video link



Observed | Reported | Resolved

01 · THE PROBLEM

Nobody Has An Independent Record Of When Someone Came To The Door

Every Medicaid-funded home-care visit in the United States must be electronically verified (EVV) with six data elements: service, recipient, caregiver, date, location, and start and end time.

In 2026 states moved from soft edits to **hard edits**: a claim without a matching verified visit is now denied automatically. Missouri started in April 2026, with Minnesota, Ohio, Indiana, Pennsylvania, and Massachusetts following.

Most agencies verify with a GPS clock-in from the caregiver's phone. It drifts, it can be spoofed, and it fails when the phone dies in a basement apartment.

Millions of client doors already have a Ring doorbell, owned by the family, producing signed events. VisitProof turns it into that missing record.

6:10 AM · SCHEDULER

Is the 6:00 AM caregiver late, at the door with no signal, or not coming at all?

TWO WEEKS LATER · COMPLIANCE

The same visit comes back as a denied claim, with nothing independent to show it happened.

EVENING · FAMILY

The client's daughter texts the agency for the third time this month: "Did anyone actually come today?"

All three share one blind spot: no independent, timestamped record of when a person arrived at and left the client's door.

02 · WHAT VISITPROOF DOES

Three Columns That Never Merge

Observed

What the door saw: arrival and departure from signed Ring events, clustered by the Presence Engine. Unobservable windows (doorbell offline) are recorded as unobservable, never as absence.

Reported

What people said: EVV clock-ins imported from the vendor, plus caregiver notes and disputes in their own signed column.

Resolved

What the agency decided: Backfill Times and Resolve are signed corrections. A correction never overwrites what the door observed.

Scheduler

Today board with late, no-show, early-departure, and unobservable exception cards. Call Caregiver, Snooze, Resolve, Backfill Times.

Compliance Lead

Reconciles every clock-in against the door: corroborated, contradicted, missing. Hash-chained, KMS-signed ledger; PDF evidence pack; Open EVV style JSON.

Family

One calm card a day, at most one alert. Granular consent, Export My Data, Delete Everything. Device Check and Live View on their own doorbell.

Caregiver

Two weeks of visits with door times next to their clock-in, so a dead phone still leaves proof they were there.

HONEST BOUNDARY

VisitProof **shows that a person arrived at the door, never who.** No facial recognition, no cross-account correlation, and it works fully with zero video. It is not an EVV system of record, not identity verification, and not resident monitoring.

Plus **Ask VisitProof** (Alexa+ track): "Did the caregiver come to Mom's today?" answered by a self-hosted MCP server (spec 2025-11-25, Streamable HTTP) in an Echo Show style simulator at /ask.

Built On The Ring Partner API, Verified Against The Live Playground

CAPABILITY	WHAT VISITPROOF DOES
Partner API calls	Typed client for users/me, devices with include (status, capabilities, configurations, location), event history, snapshots, clips, WHEP sessions, subscriptions, app integrations.
Webhooks v1.1	HMAC-SHA256 over the raw body bytes in constant time, dedupe on `meta.request_id`, 202 before any engine work.
Account linking	Token Exchange URL, Account Link URL with nonce freshness check, App Homepage URL.
Device Check	Online, motion recording, motion zone, encryption state. `4 of 4 live`
Ring history	Sync Ring History read real `on_demand` events live; epoch ms times converted to ISO; provenance `ring_history`.
Live view (WHEP)	Enforces Ring's 30 and 60 second caps; ends a connected session with no video after 8 seconds and says so.
Emulator	The Playground cannot send webhooks, so the SDK ships an HMAC-signed v1.1 emulator; every emulated event is labeled.

22 OCTOBER 2026 · LIVE PLAYGROUND

Device Check against the live Ring Partner API read **4 of 4 required checks passing**. Sync Ring History succeeded live.

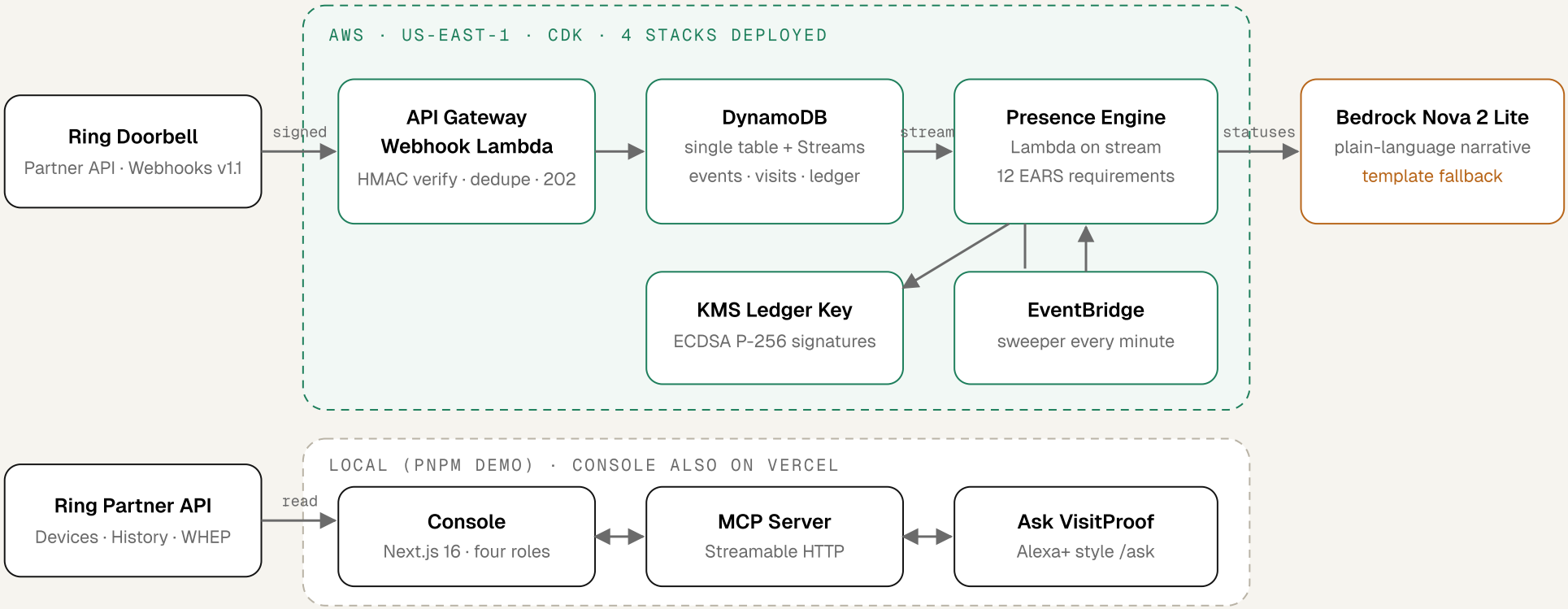
With a Motion "Simulate live view event" running in the Ring Developer Playground, VisitProof's own `/family/live` decoded **real video over WHEP: 250 frames at 1280x720**, with H264 and Opus negotiated and ICE and DTLS connected.

4/4	250	720p
DEVICE CHECKS	LIVE FRAMES	WHEP VIDEO

First contact failed honestly: strict schemas written from the docs rejected every real device payload (0 of 5 checks). We captured the responses as fixtures, fixed the schemas, and built a mock Playground so the fix stays tested after the 30 minute token expires.

Reusable: `packages/ring-partner-sdk`, an Apache-2.0 typed Ring Partner API client (Ring has no official SDK).

Signed Events In, Signed Evidence Out



Deployed And Verified

API Gateway, webhook Lambda, DynamoDB and Streams, engine Lambda, KMS ledger signing, EventBridge sweeper, SQS DLQs, CloudWatch alarms, Budgets.

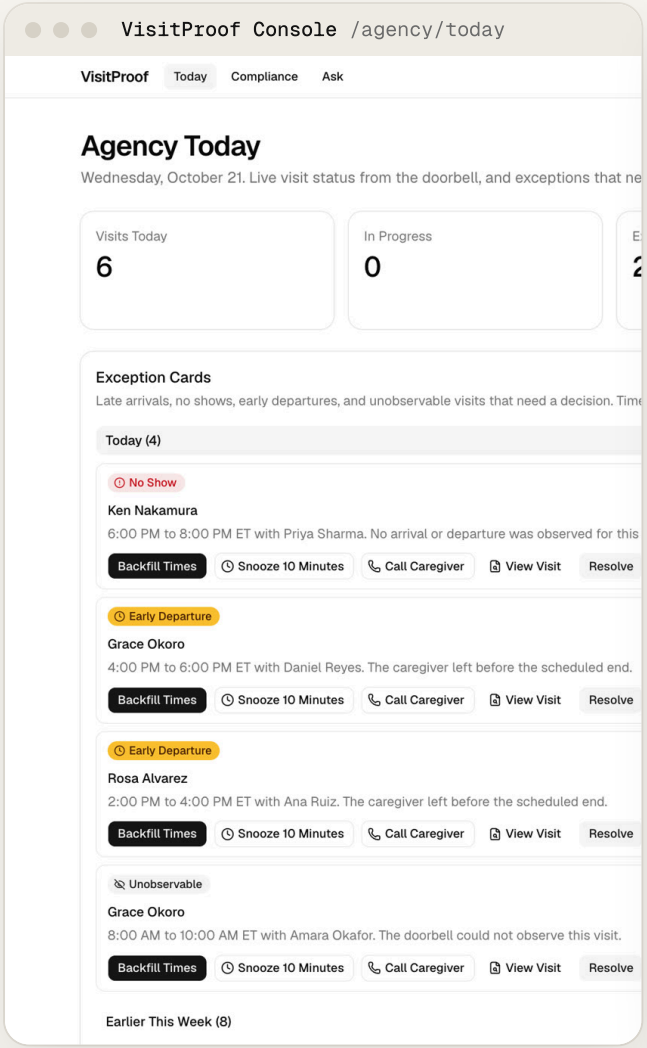
Deployed, Blocked Or Idle

Bedrock Nova 2 Lite: account quotas are 0, so calls throttle and the deterministic template is used. S3 and SNS deployed, not exercised yet.

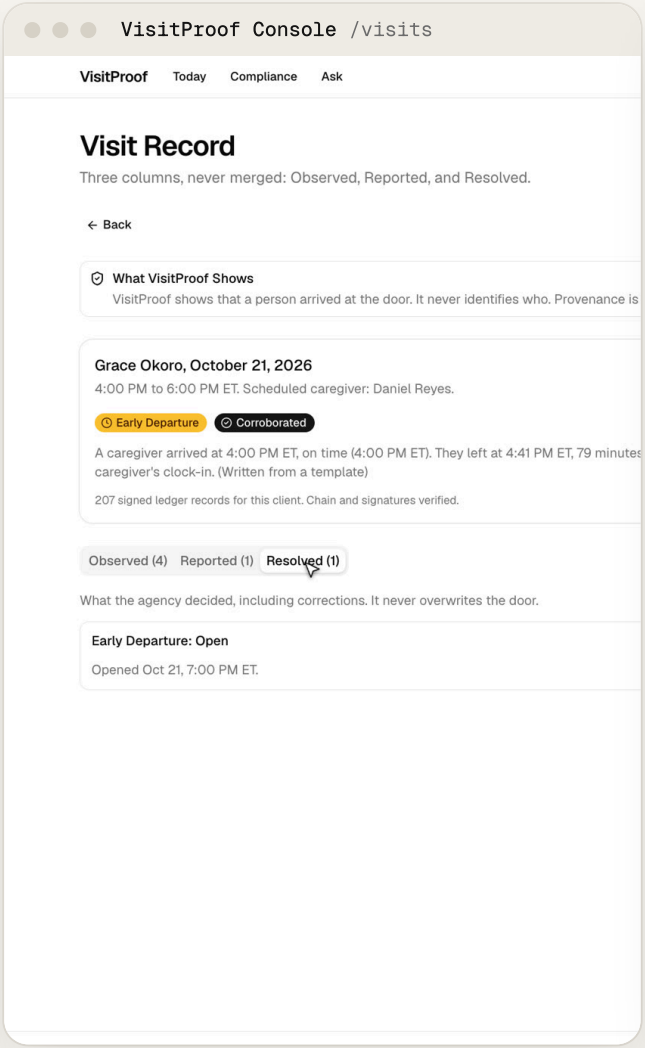
Local Or Vercel

Console (hosted on Vercel with demo data), MCP server, and Ask run locally. Strands Agents, Cognito, AgentCore, and SES are coded but not deployed.

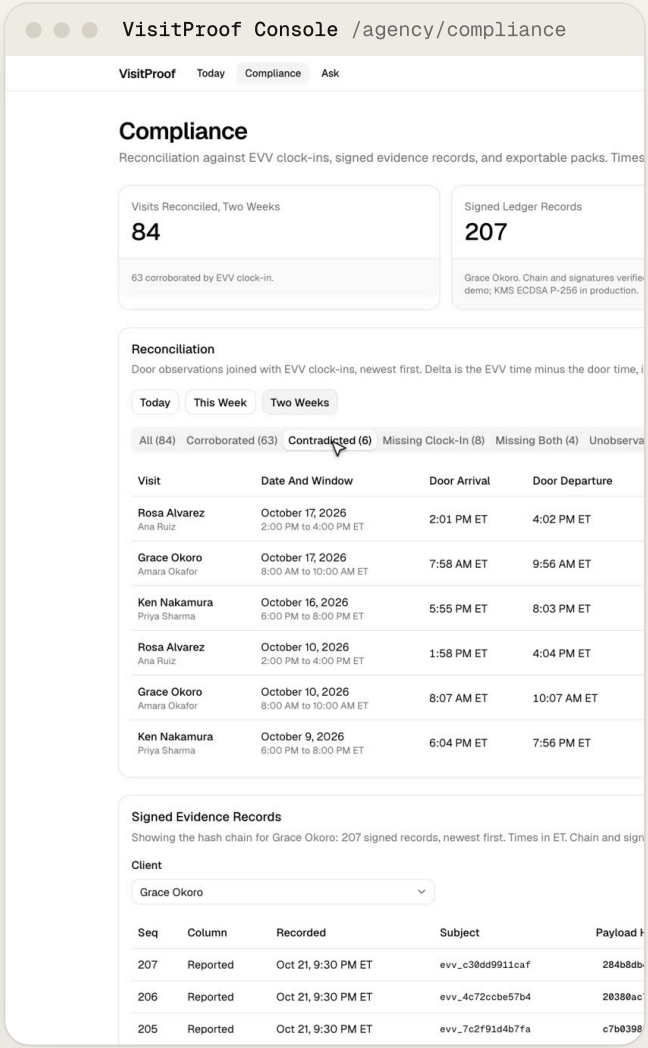
The Agency Side



Agency Today. Exception cards for late, no-show, early departure, and unobservable visits.

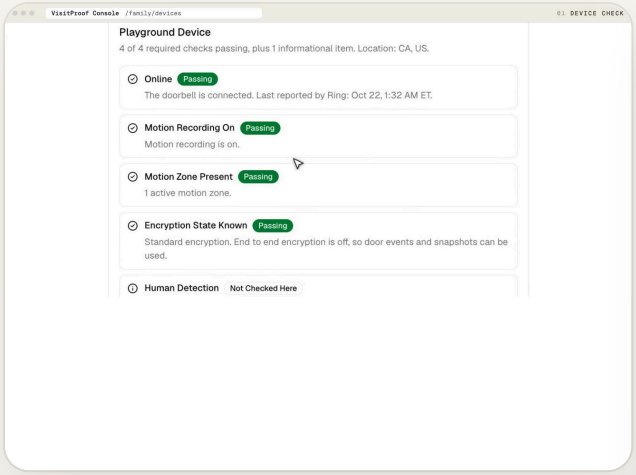


Visit Detail. Observed, Reported, and Resolved stay in separate signed columns.

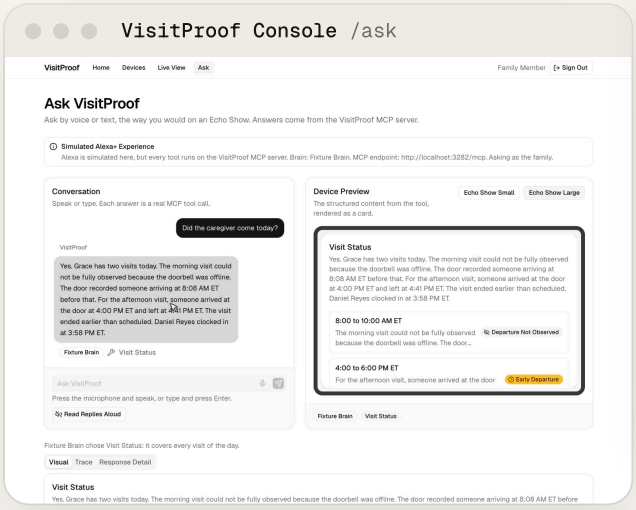


Compliance. Each EVV clock-in checked against the door; contradicted visits filtered.

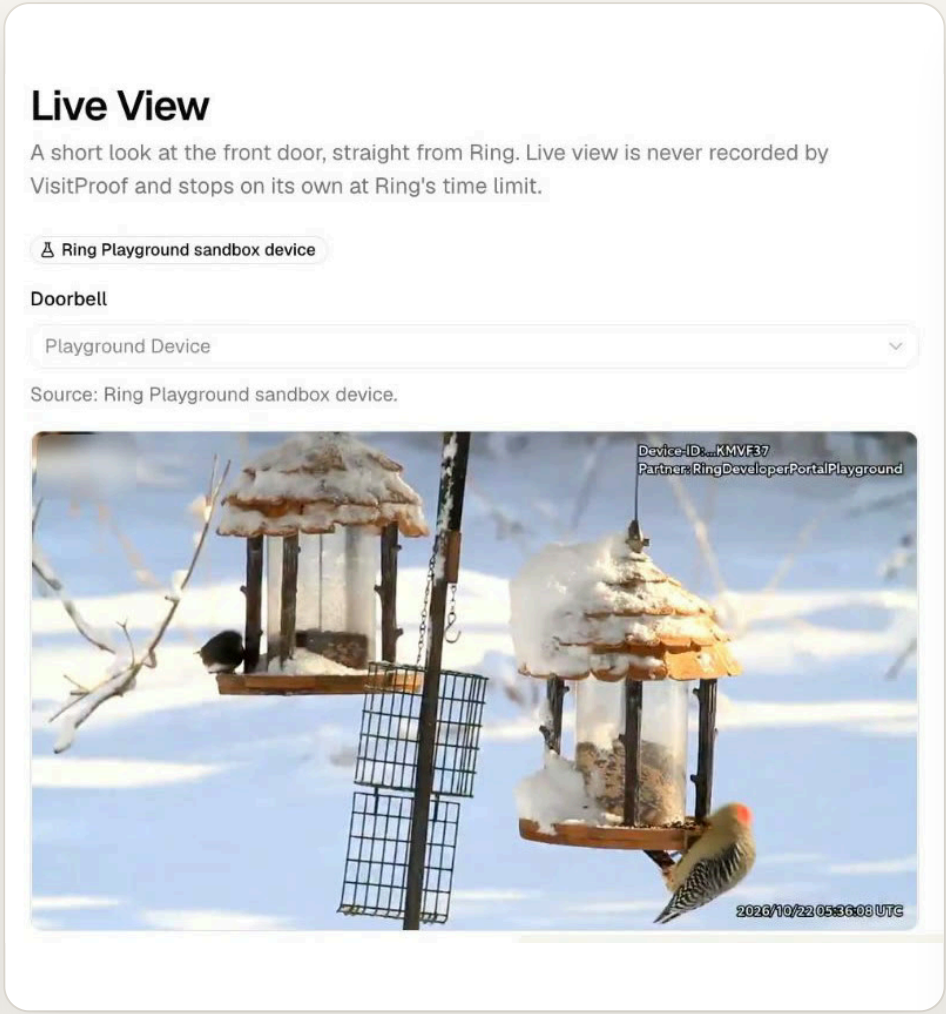
Real Ring Data, Families, And Ask



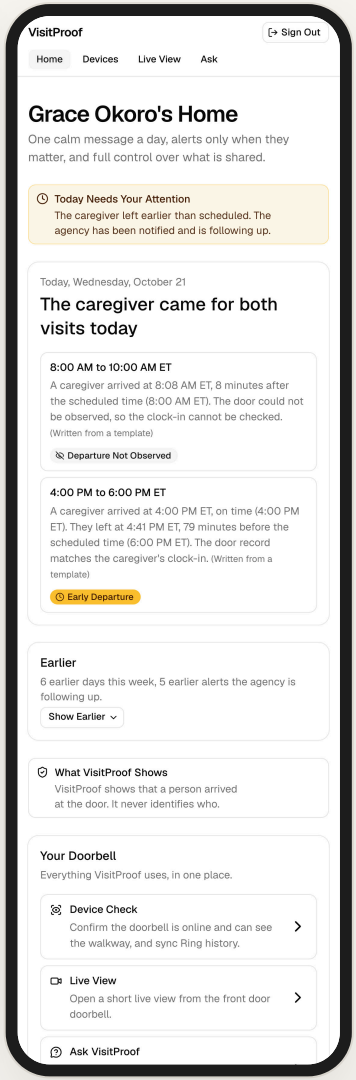
Device Check, live. 4 of 4 required checks on the Ring Playground doorbell.



Ask VisitProof. Each answer names the MCP tool it called.



Live View over WHEP, 22 October 2026. Real 1280x720 video from the Ring Playground sandbox doorbell during a simulated Motion event. Sample video: Birds on Feeders by Michael Black, CC BY 4.0.



Family. One calm card a day.

What We Checked, And How You Can Too

682	84	181
PASSING TESTS	DEMO VISITS	SIGNED WEBHOOKS

Automated Tests

Vitest, all passing on 22 October 2026: 664 across the ten workspace packages and 18 for the demo scripts. Presence Engine statement coverage is about 99 percent. GitHub Actions CI runs typecheck, lint, tests, the console build, and `cdk synth` on every push.

AWS End To End, 22 October 2026

- Signed synthetic Ring v1.1 webhook returned 202 in 815 ms; unsigned, wrong-key, and tampered requests returned 401; a repeat returned 202 duplicate.
- Door event and idempotency receipt stored in DynamoDB; the stream ran the engine with 0 errors.
- First ledger record signed by the KMS key; sweeper ran every minute with 0 errors; alarms OK, DLQs empty, spend 0 USD.

Demo Pipeline

`pnpm demo` replays two weeks through the real receiver and engine and checks the outcome against the plan: 9 late, 4 no-shows, 5 early departures, 3 unobservable, 6 contradicted, 8 missing clock-ins.

1. Live Link

Open **visitproof.codewitheugene.top** and use one-click Demo Sign In as Scheduler, Compliance Lead, Family Member, or Caregiver. It runs on demo data moved onto the current week. `/ask` and live Ring mode need a local run.

- Scheduler: open an exception card, View Visit, Backfill Times, Resolve.
- Compliance Lead: Two Weeks, filter Contradicted, download the PDF pack and JSON.
- Family Member: daily card and consent. Caregiver: a missing clock-in with door presence.

2. Local Full Demo

Node 22+ and pnpm 10+. No AWS or Ring credentials needed.

```
git clone https://github.com/CodeWithEugene/Amazon-Developer-Hackathon.git
cd Amazon-Developer-Hackathon && pnpm install && pnpm demo
```

When it prints "VisitProof demo is ready", open localhost:3000. This also starts the MCP server for `/ask`.

3. Live Ring Mode

Mint a 30 minute token in the Ring Developer Playground, save it with `pnpm ring:token`, and run the console. Device Check and Sync Ring History read the sandbox doorbell. For video, start a Motion, Package, or Vehicle "Simulate live view event" first, then Start Live View on `/family/live`.

What We Hit, And What Would Fix It

#	FRICTION	SEVERITY
1	Playground cannot deliver webhooks or produce motion and doorbell events; history is empty. We built a signed emulator.	High
16	Real device payloads use undocumented type names and many nulls; strict schemas rejected all four sub-resources.	High
18	Live view connects but sends no video unless a Playground simulation is running. Resolved: 250 frames at 1280x720.	High
4	`ring-api-helloworld` checks a static bearer token instead of the `X-Signature` HMAC.	High
2	Playground tokens last about 30 minutes with no refresh.	Medium
17	Subscriptions returns 422 for the Playground user.	Low
9	Alexa+ docs disagree on the MCP protocol version (2025-03-26 vs 2025-11-25).	Medium

Top Feature Requests

1. **Signed test webhooks** from the Playground to a developer URL.
2. **Response examples** with real type names and every nullable field.
3. **A sandbox doorbell that streams**, or a note next to the WHEP docs that video needs a simulation.
4. **Official test vectors** for webhook signatures and nonces, and a sample that verifies them.
5. **An OpenAPI or JSON schema** for the Partner API.
6. Pre-seeded history events, TAKE mode next to `e2e_encryption`, longer developer tokens.

What Worked Well

Clean JSON:API, fast 200s, one devices call with include for all of Device Check, a token in seconds, WHEP signaling first time, and HMAC over raw bytes with a request id. The read-only scope makes it easier for families to say yes.

Full log, 18 entries with task, steps, expected, actual, severity, workaround, and suggestion: `github.com/CodeWithEugene/Amazon-Developer-Hackathon/blob/main/docs/friction-log.md`

What Is Next, And What We Do Not Claim

Roadmap

- Private-app pilots with home-care agencies.
- Ring Appstore certification (questionnaire drafted in `docs/compliance.md`).
- Register the deployed webhook URL with Ring so real events reach AWS.
- Raise Bedrock Nova 2 Lite quotas for model narratives; enable SES digests and alerts.
- Cognito sign-in and the MCP server on AgentCore Runtime.
- Alternate-EVV vendor onboarding in Provider Choice states.
- An Echo Show family view through MCP Apps.

Honest Limitations

- No real Ring webhook traffic yet: the Playground cannot send webhooks, so the event path uses a labeled, signed emulator.
- The live Ring run used the shared Playground sandbox doorbell, not a family's doorbell.
- Bedrock narratives are throttled (quota 0); the deterministic template is used.
- The hosted console runs on demo data; `/ask` and live Ring mode need a local run.
- S3 and SNS are deployed but not exercised; Strands Agents, Cognito, AgentCore, and SES are coded but not deployed.

BOUNDARIES Not an EVV system of record, not identity verification, not resident monitoring, and not a medical device. No facial recognition. Never correlates across Ring accounts. Works fully without video.



VisitProof shows that a person arrived at the door. It never identifies who.

`visitproof.codewitheugene.top · github.com/CodeWithEugene/Amazon-Developer-Hackathon · Apache-2.0`